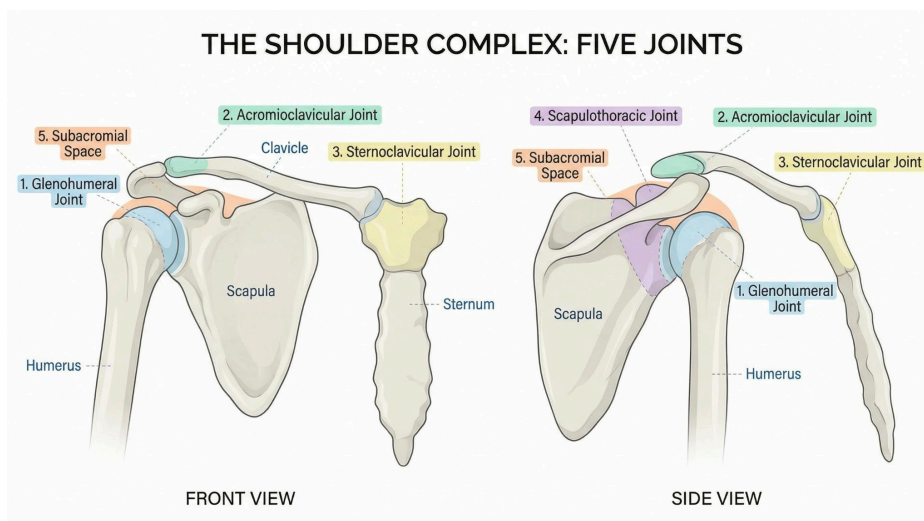


Anatomy of the Shoulder Joint Complex

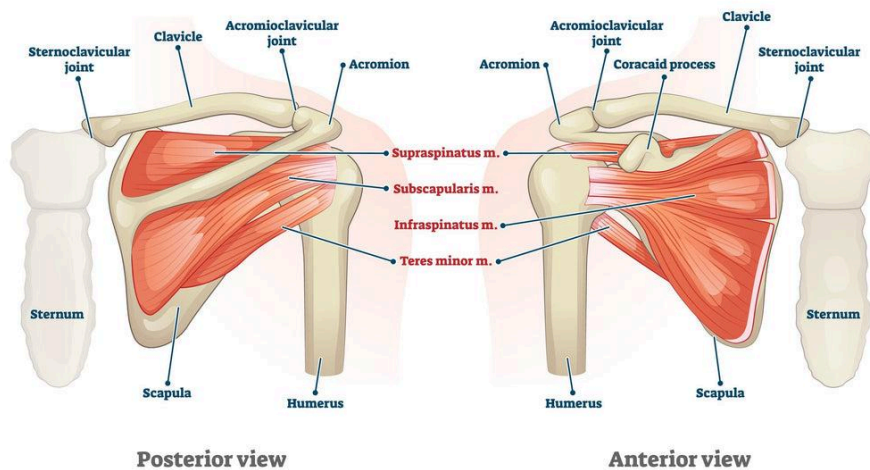
1. Osteology (Bony Structure)

The shoulder complex is composed of three primary bones and specific landmarks that form the protective "roof" and socket of the arm.

- Humerus (Upper Arm Bone): The proximal end features a smooth, hemispherical structure called the humeral head (the "ball" of the joint).
- Scapula (Shoulder Blade): A flat, triangular bone on the upper back containing three critical landmarks:
 - Glenoid: A naturally flat, shallow socket located on the lateral edge of the scapula that receives the humeral head.
 - Acromion: A rigid bony projection extending forward from the scapular spine to form the protective functional roof of the shoulder.
 - Coracoid Process: A hook-like bony knob that projects forward from the front of the scapula near the joint line.
- Clavicle (Collarbone): An S-shaped strut that bridges the arm assembly to the central skeleton.



SHOULDER ANATOMY



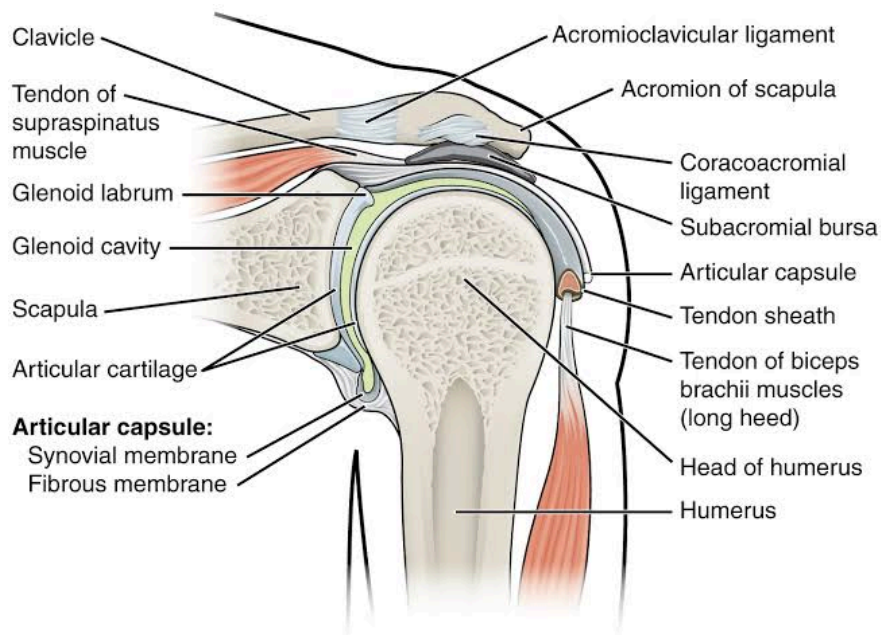
2. Joint Articulations & Cartilage

The shoulder complex is formed by four distinct joints working in unison to position the hand in space.

- Glenohumeral Joint: The primary ball-and-socket joint where the humeral head meets the shallow glenoid socket.
- Acromioclavicular (AC) Joint: A planar gliding joint located where the outer end of the clavicle articulates with the acromion of the scapula.
- Sternoclavicular (SC) Joint: The anchor point on the front chest connecting the inner clavicle to the sternum (breastbone). This is the only true bony connection linking the upper extremity to the main skeleton.
- Scapulothoracic Joint: A physiological "false joint" where the scapula glides across the rib cage. It relies entirely on surrounding muscles to keep the socket aligned during arm movement.

Articular Cartilage: A rubbery, highly slippery material covering the ends of the bones in these joints. Because the shoulder is non-weight-bearing, it is slightly

thinner than in joints like the hip or knee, but serves the same critical functions: absorbing shock and minimizing structural friction.



3. Myology (Muscles and Tendons)

Muscles provide the active dynamic stability and mechanical power required to generate force and direct shoulder movement.

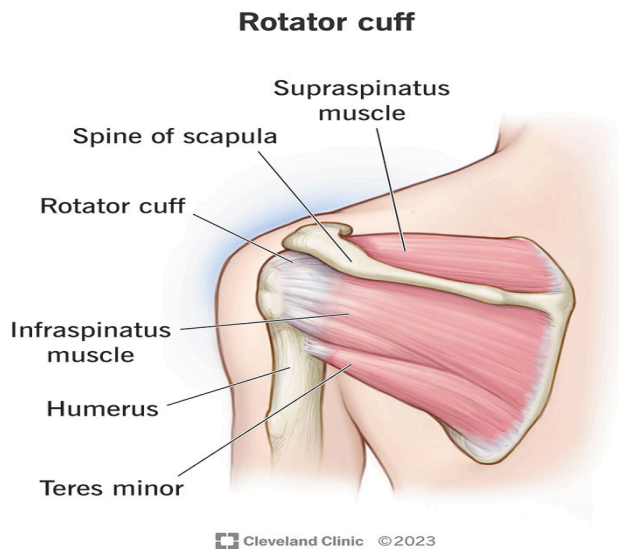
The Rotator Cuff (Deep Stabilizers)

A deep layer of four distinct muscles lying just outside the joint capsule. As they approach the upper humerus, their individual tendons fuse into a single structural sheet called the rotator cuff tendon. They follow the SITS acronym:

1. **Supraspinatus:** Sits in the top groove of the shoulder blade, passing beneath the acromion. It initiates the first 15 degrees of lifting your arm out to the side (abduction).
 2. **Infraspinatus:** Covers the back surface of the shoulder blade, providing external (outward) rotation.
 3. **Teres Minor:** A small, narrow muscle running beneath the infraspinatus that assists in external rotation.
 4. **Subscapularis:** Located on the *front* face of the shoulder blade (facing the ribs). It wraps forward to attach to the lesser tubercle, driving internal (inward) rotation.
- **Primary Function:** The rotator cuff coordinates contraction to hold the humeral head tightly into the shallow glenoid socket, keeping the joint centered as the larger arm muscles move it.

The Deltoid Muscle (Outer Power Mover)

The large, powerful outer layer forming the physical shape of your shoulder. The deltoid muscle provides the primary mechanical force required to lift the arm once it is initiated away from the side of the body.



4. Neurovascular Structures & Friction Management

- The Subacromial Bursa: A watertight, fluid-filled sac sandwiched between the superficial deltoid muscle and the deep rotator cuff tendons. It contains lubricating fluid to protect the sliding surfaces and reduce friction where these two distinct tissue layers slide against each other. (If the rotator cuff fails to keep the joint centered, the humeral head glides upward, squeezing this bursa against the hard acromion, causing Shoulder Impingement).
- The Brachial Plexus Nerves: Major nerves traveling from the neck down into the arm through the axilla (armpit) under the shoulder. Three main nerves—the radial, ulnar, and median nerves—transmit motor control and sensory signals.
- Axillary Nerve: A specific nerve branch traveling around the back of the joint to supply sensation to the outer shoulder skin and motor signals to the deltoid muscle.
- Axillary Artery: A large, high-volume blood vessel passing through the armpit alongside the nerve paths. It branches into smaller vessels, giving the shoulder a rich blood supply.

